# Mark Julius P. Mella

Software Engineer | Agentic AI Engineer

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## Professional Summary

Software engineer with 6+ years across freelance and product development. Plan and build web applications with full-stack knowledge, REST APIs and persistent data. Focus on frontend architecture, API integration, performance and reliability; transitioning into agentic AI engineering.

## Technical Skills

**Programming and frontend:** JavaScript, TypeScript, Vue 2 / Vue 3, React, Next.js, HTML, CSS, Sass, Tailwind CSS

**APIs and state management:** Vue Router, Pinia, Vuex, REST APIs, Axios, TanStack Vue Query, SignalR, Zustand, Zod, React Hook Form, Mobile WebViews

**Performance and SEO:** Caching, IndexedDB, Data fetching, Image optimization, Lazy loading, Code splitting, Hydration, Lighthouse, Search engine optimization (SEO)

**Agentic AI workflows:** Cursor, Claude Code, Codex, ChatGPT, AI-assisted research, Structured prompts, Task decomposition, Context and constraints, Output validation, Automation knowledge

**Backend, data and development tools:** Node.js, PHP, Laravel, Firebase, Firestore, Chart.js, Vite, Git, Figma-to-code, Agile / Scrum

## Work Experience

### Software Developer — Frontend & API Integration

May 2023 - Present

DigiPlus Interactive Corp

- Integrate REST APIs and real-time services for ArenaPlus sportsbook features, including sports data, accounts, deposits, withdrawals, and customer support.
- Develop and maintain sportsbook and SEO-focused web applications using Vue 3, TypeScript, Vite, and Tailwind CSS.
- Build shared layouts, reusable components and design tokens for responsive light and dark interfaces.
- Optimize startup loading through caching, image and font delivery, lazy loading, and code splitting.
- Support Vue 2-to-Vue 3 migrations and collaborate with product managers, designers, and backend engineers.

### Frontend Developer

May 2022 - May 2023

Solid Leisure Solutions

- Maintained and enhanced BingoPlus web interfaces for a digital bingo platform.
- Translated Figma designs into responsive user interfaces.
- Integrated REST APIs to connect interfaces with backend services.
- Integrated customer-support chat into web applications.
- Optimized frontend performance to improve application responsiveness.

## Education

### Bachelor of Science in Computer Science

2018 - 2022

Pamantasan ng Lungsod ng Pasig

## Projects

### Application Foundations, API Integration and Reliability

Scaffolded authenticated routing, layouts, stores and services; integrated account and wallet APIs. Built shared themes, deferred SDK loading, and refined caching, stale-request handling and session cleanup.

Vue 3, TypeScript, REST APIs, Pinia, Vite

### Product Storefront and Inventory Management

Contributed image loading, description fallbacks and version-aware caching that retains data on refresh failure. Added subcategory management, Firestore access rules, publication controls and inventory cache refresh.

React, Next.js, TypeScript, Firebase, IndexedDB

### Catalog Editing Workspace

Contributed multi-catalog management and saved design settings. Persisted item ordering in Firestore with optimistic updates and rollback when writes fail.

React, Next.js, TypeScript, Zustand, Zod, Firestore

### Campaign Forms, Event Draws and Reporting

Adjusted receipt upload and participation states; updated draw participant loading and animation. Added duplicate receipt review, date filters and reporting views.

React, Next.js, Vue 3, TypeScript, Firebase, Chart.js

## Agentic AI Workflows and Automation

My foundations began with self-study in college in 2018. I started learning AI in early March 2025 and am now transitioning into agentic AI engineering.

- Use Codex and ChatGPT for research, and Cursor and Claude Code for code audits, debugging, and incremental refactoring.
- Define scope, context, constraints and acceptance criteria; break AI-assisted changes into reviewable phases.
- Review generated changes and affected call sites; validate behavior through code review, type checks, builds and browser inspection.

## Additional Work Experience

### Frontend Developer - Internship

March - April 2022

Public Safety Savings and Loan Association, Inc. (PSSLAI)

- Built employee risk-calculation interfaces for a risk management application using React, Node.js, and AWS.

### Freelance Frontend Developer

2020 - 2022

- Translated Figma designs into responsive web and mobile layouts, and built reusable UI components.

### Freelance Computer Technician

2020 - 2022

- Installed and configured hardware and software; diagnosed and repaired desktop and laptop issues.

## Awards and Certifications

- Best in Thesis - AWIT, an interpreted Turing-complete scripting language written in C with bytecode generation and garbage collection.
- Dean's Lister, 2021-2022.
- Google Tech Challenge certificate, September 28, 2019.